The Laryngoscope
Lippincott Williams & Wilkins
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Vocal Fold Immobility: A Longitudinal Analysis of Etiology Over 20 Years

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Objectives: To determine the current etiology of vocal fold immobility, identify changing trends over the last 20 years, and compare results to historical reports.

Study Design: The present study is a retrospective analysis of all patients seen within a tertiary care institution between 1996 and 2005 with vocal fold immobility. The results were combined with a previous study of patients within the same institution from 1985 through 1995. Results were compared to the literature.

Methods: The medical records of all patients assigned a primary or additional diagnostic code for vocal cord paralysis were obtained from the electronic database.

Results: Eight hundred twenty-seven patients were available for analysis (435 from the most recent cohort), which is substantially larger than any reported series to date. Vocal fold immobility was most commonly associated with a surgical procedure (37%). Nonthyroid surgeries (66%), such as anterior cervical approaches to the spine and carotid endarterectomies, have surpassed thyroid surgery (33%) as the most common iatrogenic causes. These data represent a change from historical figures in which extralaryngeal malignancies were considered the major cause of unilateral immobility. Thyroidectomy continues to cause the majority (80%) of iatrogenic bilateral vocal fold immobility and 30% of all bilateral immobility.

Conclusions: This 20-year longitudinal assessment revealed that the etiology of unilateral vocal fold immobility has changed such that there has been a shift from extralaryngeal malignancies to nonthyroid surgical procedures as the major cause. Thyroid surgery remains the most common cause of bilateral vocal fold immobility.

Key Words: Vocal fold paresis, vocal fold paralysis, hoarseness, unilateral, bilateral.

***Laryngoscope,* 117:1864–1870, 2007**

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Editor's Note: This Manuscript was accepted for publication May 8, 2007.

Presented at the Triological Society Combined Sections Meeting, Marco Island, Florida, U.S.A., February 16, 2007.

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DOI: 10.1097/MLG.0b013e3180de4d49

INTRODUCTION

Vocal fold immobility is a broad term used to describe vocal folds that are restricted secondary to mechanical fixation or neuropathy. Mechanical fixation may result from an arytenoid dislocation, edema or inflammation of the glottis, or neoplastic invasion. Neurogenic immobility may occur with lesions in the motor cortex or compromise of the recurrent laryngeal nerve at any point along its course from the jugular foramen to the carotid sheath, mediastinum, and either around the subclavian artery on the right or the aortic arch on the left, to the tracheoesophageal groove. Vocal fold movement may be partially limited, described as hypomobility, or completely fixed or immobile. Vocal fold immobility may be unilateral or bilateral. The distinction between hypomobile or immobile and unilateral or bilateral is important in workup and management because of the implications that these factors have on the potential etiology and prognosis. Unilateral immobility may present as dysphonia or dysphagia, or be asymptomatic. The mobile fold may compensate for contralateral dysfunction. Bilateral immobility may present as obstruction often requiring intubation and tracheotomy. Many patients with both unilateral and bilateral immobility may have spontaneous resolution, depending on the etiology. However, neoplastic invasion and iatrogenic division of the recurrent laryngeal nerve result in permanent immobility. Inflammation, infection, or iatrogenic stretching of the recurrent laryngeal nerve may cause only temporary immobility. If there is no recovery, patients with unilateral immobility may benefit from a medialization procedure, temporary or permanent. Select patients with bilateral immobility may benefit from arytenoidectomy, partial cordectomy, or lateralization. The etiology of unilateral and bilateral immobility, therefore, has a profound influence on workup, timing, and method of management.[1]

Previous studies of the etiology of unilateral vocal fold immobility have suggested multiple etiologies. Intubation,[2] lung cancer,[3] idiopathic,[4] and nonthyroid surgeries[5] have all been reported as the single most common cause. Other common causes reported include thyroid surgery, non-lung malignancies, and central nervous system disease. Surgical iatrogenic injury has historically been the most common cause of bilateral immobility.[3,4] As the differential diagnosis is extensive, the workup may be

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extensive as well.[1] A previous retrospective review of the etiology of vocal fold immobility of 397 cases from the Henry Ford Health System from 1985 to 1995 showed growing percentages of extralaryngeal malignancies and surgery-related injuries.[6] In this study, cases of vocal fold immobility from within the same institution over the next 10 years, from 1996 through 2005, are reviewed. The study allows evaluation of longitudinal trends within a single institution over the last 20 years and is compared to studies performed in many institutions over the last 50 years.

MATERIALS AND METHODS

The medical records of all patients assigned a primary, secondary, or tertiary diagnostic code for vocal fold paralysis (ICD-9 478.3) within the Henry Ford Health System from 1996 through 2005 were obtained from the electronic database. This diagnostic code captures all patients with vocal fold hypomobility and immobility. Each patient chart was retrospectively reviewed, with institutional review board approval, to determine the following: gender; age at diagnosis; side of vocal fold immobility; method of diagnosis of vocal fold immobility; additional studies for work-up, including chest radiograph (CXR), computed tomography (CT), magnetic resonance imaging (MRI), barium swallow, and electromyogram (EMG); histopathology; and the etiology of the immobility. All cases of vocal fold immobility were diagnosed using indirect laryngoscopy, flexible fiberoptic laryngoscopy, or video stroboscopy. The etiology, including whether it was idiopathic, was recorded if adequately documented by the physician or the result of directed studies. Records were excluded for inadequate documentation of vocal fold immobility or its etiology; initial presentation prior to 1996, because the record would have been included in the prior study; a child less than 14 years of age; and a primary malignancy of the upper respiratory tract, such as a squamous cell carcinoma of the true vocal fold. Microsoft Excel Pivot Tables (Microsoft Corp, Redmond, WA) were used to tabulate the results. The Jonckheere-Terpstra exact test was used to evaluate trends in proportions across years.

RESULTS

A diagnostic code of vocal fold immobility was assigned to 808 patients. Of those, 373 patients were excluded, 26 for prior presentation within the institution, 18 for age less than 14 years old, 26 for an upper respiratory tract malignancy, 268 for normal vocal fold mobility, 30 for inadequate information available for analysis, and 5 for inadequate documentation of etiology. Many patients with inadequate documentation presented to the emergency room or the office, but had no follow-up to confirm the etiology. A total of 435 patients were available for analysis of the etiology of vocal fold immobility.

Of the 435 patients, 363 (83%) were unilateral and 72 (17%) were bilateral. The average age was 60 years, ranging from 14 to 93. There were 256 female patients (61%) and 179 male patients (39%). For patients of unilateral immobility, 58% of these were female and 42% were male. There were 142 cases of the right vocal fold (39%) and 221 were of the left vocal fold (61%). For patients of bilateral immobility, 61% were female and 39% were male.

The etiology of unilateral vocal fold immobility for the most recent cohort is shown in Figure 1. The most common causes were surgical iatrogenic injury, 168 (46.3%); followed by idiopathic, 64 (17.6%); and non-upper respiratory malignancies, 48 (13.5%). Lung malignancies, excluding metastatic lung disease to mediastinal or paratracheal lymph nodes, were the most common malignant causes, accounting for 24 cases (6.6%). There were many other rare causes of unilateral vocal fold immobility. Rare tumors responsible included two esophageal cancers, acute myelogenous leukemia, glossopharyngeal and vagus nerve palsies from a cervical spine mass at levels C1 and C2, and four benign tumors of the vagus nerve. One patient had an unnamed neuropathy documented as the etiology. Another had an undetermined neuropathy and was later found to have a brain stem lymphoma. In one case it was unknown if neck trauma or the subsequent neck exploration caused vocal fold immobility.

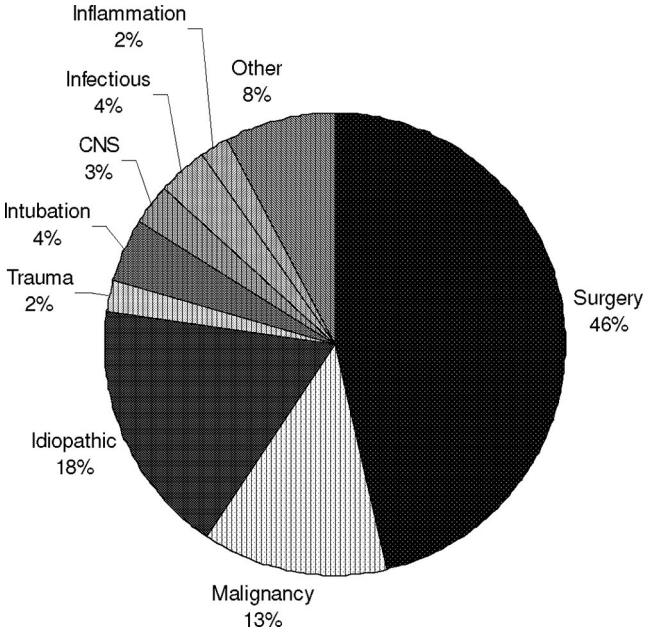

Fig. 1. Etiology of unilateral vocal fold immobility, 1996 to 2005.

Nonthyroid surgeries (111 [30.6% of all causes of immobility]) caused twice as many cases of unilateral vocal fold immobility as thyroid surgeries (57 [15.7%]). Thyroid surgeries (33% of all surgical causes) include thyroidectomy (26%), parathyroidectomy (6%), and both thyroidectomy and parathyroidectomy performed during the same surgery (1%). The surgical procedures most likely to cause unilateral vocal fold immobility are shown in Figure 2. The most common nonthyroid surgery to cause vocal fold immobility was anterior cervical spine corpectomy and fusion (15%). The next most common surgical cause of unilateral vocal fold immobility was carotid endarterectomy (11%). Other iatrogenic causes include surgery of the lungs, mediastinum, heart, heart valves, aorta, and skull base; craniotomies for tumor or aneurysm; surgery of the esophagus; and other procedures of the neck. There were cases of unilateral vocal fold immobility after a cricopharyngeal myotomy, botox injection for cricopharyngeal spasm, esophageal dilation, and excision of benign vocal fold lesions.

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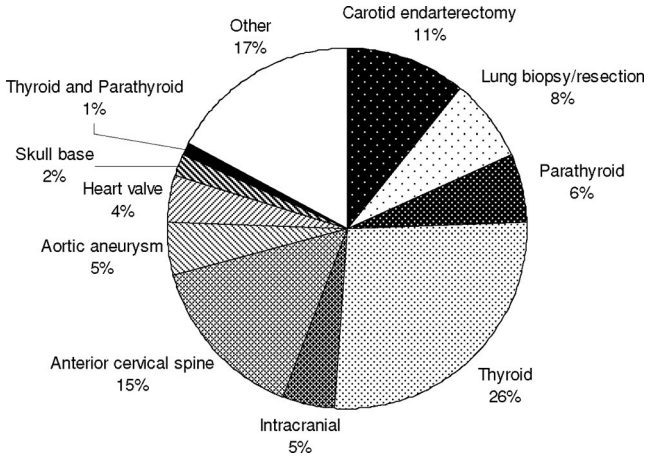

Fig. 2. Surgical causes of unilateral vocal fold immobility, 1996 to 2005.

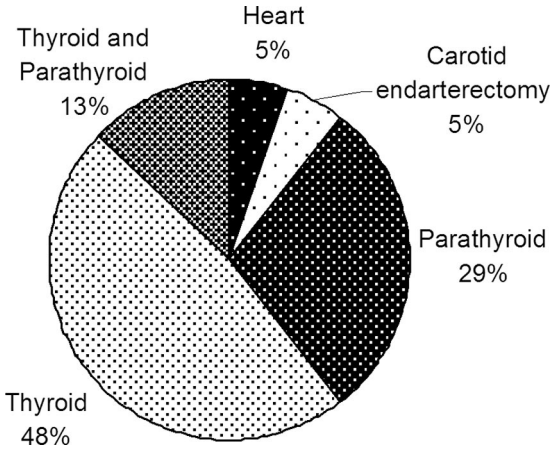

Fig. 4. Surgical causes of bilateral vocal fold immobility, 1996 to 2005.

The etiology of bilateral vocal fold immobility for the most recent cohort is shown in Figure 3. Similar to unilateral immobility, surgery was the most common cause of immobility (55.6% [40 of 72 patients]). For surgical etiologies of bilateral immobility, however, thyroid surgeries (thyroid, parathyroid, or both) resulted in the majority of cases (89%). The surgical causes of bilateral immobility in the most recent cohort are shown in Figure 4. Malignancy caused bilateral immobility in seven patients (9.7%). There were cases of pulmonary metastases to mediastinal nodes, one case of an esophageal malignancy that eroded through the trachea, one case of cervical cancer metastasizing to the paratracheal nodes, and one case of lymphoma in the mediastinum. Intubation also caused immobility in seven patients (9.7%). After surgery, malignancy, and intubation, the next most common causes of bilateral immobility are idiopathic (6 [8.3%]) and neuropathy (5 [6.9%]), such as Myasthenia Gravis or Kennedy's syndrome.

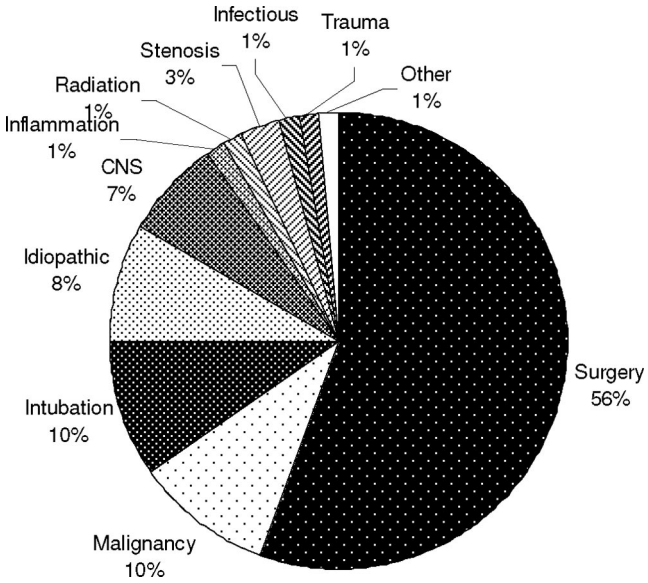

Fig. 3. Etiology of bilateral vocal fold immobility, 1996 to 2005.

The prior study from 1985 to 1995 within the same institution had a total of 280 patients (70%) with unilateral immobility and 117 (30%) with bilateral. The total number of patients included in the longitudinal study is 827; 643 (78%) with unilateral immobility and 184 (22%) with bilateral immobility. The combined data for the etiology of unilateral immobility is presented in Table I. Surgery was the most common cause (36.5%). Thyroid surgery accounted for one third of surgical causes (12.4%

TABLE I.
Etiology of Unilateral Vocal Fold Immobility, 1985 to 2005.

	1985–1995		1996–2005		1985–2005	
	n	%	n	%	n	%
Surgery	67	23.9	168	46.3	235	36.5
Thyroid	23	8.2	57	15.7	80	12.4
Nonthyroid	44	15.7	111	30.6	155	24.1
Malignancy	69	24.6	49	13.5	118	18.4
Lung	55	19.6	24	6.6	79	12.3
Metastatic	4	1.4	12	3.3	16	2.5
Thyroid	3	1.1	8	2.2	11	1.7
Esophageal	7	2.5	2	0.6	9	1.4
Other	0	0	3	0.8	3	0.5
Idiopathic	55	19.6	64	17.6	119	18.5
Trauma	31	11.1	8	2.2	39	6.1
Intubation	21	7.5	16	4.4	37	5.8
CNS	22	7.9	11	3.0	33	5.1
Infectious			13	3.6	13	2.0
Inflammation			7	1.9	7	1.1
Radiation			3	0.8	3	0.5
Stenosis			3	0.8	3	0.5
Aortic aneurysm			2	0.6	2	0.3
Other			19	5.2	19	3.0
Total	280		363		643	

CNS = central nervous system.

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TABLE II.
Etiology of Bilateral Vocal Fold Immobility.

	1985–1995		1995–2005		1985–2005	
	n	%	n	%	n	%
Surgery	30	25.7	40	55.6	70	37.0
Thyroid	21	18	35	48.6	56	26.9
Nonthyroid	9	7.7	5	6.9	14	7.4
Malignancy	20	17	7	9.7	27	14.3
Lung	6	5.1	3	4.2	9	4.8
Metastatic	4	3.4	2	2.8	6	3.2
Thyroid				0.0	0	0.0
Esophageal	10	8.5	1	1.4	11	5.8
Other			1	1.4	1	0.5
Intubation	18	25.4	7	9.7	25	13.2
Idiopathic	15	12.8	6	8.3	21	11.1
CNS/Neuropathy	15	12.8	5	6.9	20	10.6
Trauma	13	11.1	1	1.4	14	7.4
RA/Inflammation	4	4.3	1	1.4	5	2.6
Radiation	2	1.7	1	1.4	3	1.6
Stenosis			2	2.8	2	1.1
Infectious			1	1.4	1	0.5
Other			1	1.4	1	0.5
Total	117		72		189	

CNS = central nervous system; RA = rheumatoid arthritis.

of all unilateral vocal fold immobility). Nonthyroid surgeries were responsible for two thirds of surgical causes (24.1% of the total). Idiopathic and non-upper respiratory malignancies were the second and third most common causes (18.4% and 18.5%, respectively). Two thirds of malignancies were of the lung (12.3% of all cases of unilateral immobility). Trauma (6.1%), intubation (5.8%), central nervous system disease (5.1%), infection (2.0%), inflammation (1.1%), radiation therapy (0.5%), stenosis (0.5%), and aortic aneurysm (0.3%) were less common causes.

The combined longitudinal data for the etiology of bilateral immobility over the last 20 years are presented in Table II. As with unilateral immobility, surgery (37.0%) far exceeds malignancy (14.3%) as the most common cause of bilateral immobility. Eighty percent of the surgeries were of the thyroid, parathyroid, or both. Most malignancies were either lung (4.8%), esophageal (5.8%), or metastatic (3.2%). Less common causes of bilateral immobility include intubation (13.2%), idiopathic (11.1%), central nervous system or peripheral nervous system disease (10.6%), and trauma (7.4%).

DISCUSSION

This study yields both expected and unexpected trends in the etiology of vocal fold immobility, both within this institution and compared to prior literature. The combined data from 1985 to 2005 in this report comprise the largest study of unilateral vocal fold immobility reported.[3,7,8] Although other studies have spanned 20 years,[9] this is the only reported longitudinal study within one institution.

Comparing the current study to the study from the previous 10 years within the same institution, there are significant differences. For unilateral vocal fold immobility, the proportion of cases caused by surgical injury doubled from 23.9% to 46.4% (Table I). Both thyroid and nonthyroid surgeries doubled. There was an increase in the total number of thyroid surgeries performed over that time, which may account for this increase. The proportion of malignant etiologies decreased 50% (from 24.7% to 13.2%), despite an increase in the number of metastatic and thyroid malignancies (1.4% to 3.3% and 1.1% to 1.2%, respectively). The number of idiopathic cases has remained stable, and now equals malignancy as the second most common cause of unilateral vocal fold immobility. The number of traumatic injuries is significantly less, as is the number of intubation-related causes. This may be attributed to changes in management of intubated patients within the institution, in particular, well-established hospital guidelines regarding length of intubation and endotracheal and tracheostomy care. These trends in increasing immobility from thyroid surgery and nonthyroid surgery and decreasing immobility from malignancy are consistent with the literature, as shown in Table III.

Table III shows the changing etiology of unilateral vocal fold immobility over the last century. There is value in estimating whether a trend exists in these data. However, to aid a relatively subjective evaluation of the data, the Jonckheere-Terpstra exact test[10] was used as an additional tool to determine whether there was a statistically significant increasing or decreasing trend based on the values reported in order of their year of publication. There is no great statistical test to evaluate trends. In addition, this test may be limited as the trend is determined by publication date regardless of how many years passed between publication dates or the time period over which vocal fold immobility data were gathered. This information was not published in all reports.

There is a statistically significant decrease in malignant causes of unilateral vocal fold immobility in the last 10 years, from 34.8% lung and nonlung malignant etiologies published in 1970 by Parnell and Brandenburg,[2] to 13.5% in the current study. Malignancy may become a less significant cause of vocal fold immobility as lung cancers are detected and treated earlier. There is a statistically significant growing trend in surgery-related injuries to the recurrent laryngeal nerve. The proportion of nonthyroid surgeries causing vocal fold immobility has increased, from 19% of surgeries (4.6% nonthyroid of 24.4% total surgical cases) published in 1970 by Parnell and Brandenburg,[2] to 66% (30.6% nonthyroid of 46.3% total surgical cases) in the current study. The increase in proportion of surgical injuries may be secondary to an increase in the numbers of surgeries, including thyroidectomy, anterior approaches to the cervical spine, carotid endarterectomies, or procedures elsewhere along the course of the recurrent laryngeal nerve. Nearly all other causes of unilateral vocal fold immobility show declining proportions, including central nervous system causes, aortic or cardiac, and traumatic. These also may be relative changes as surgical cases have increased.

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TABLE III.
Changing Etiology of Unilateral Vocal Fold Immobility.

	Parnell 1970, n = 86 (%)	Maisel 1974, n = 127 (%)	Titche 1976, n = 134 (%)	Terris 1992, n = 84 (%)	Benninger 1985–1995, n = 280 (%)	Henry Ford Hospital 1996–2005, n = 363 (%)	Trend *P* Value
Thyroid surgery	19.8	8.7	3.7	8.3	8.2	15.7	Increasing* .0496
Nonthyroid surgery	4.6	7.0	6.7	26.2	15.7	30.6	Increasing* <.0001
Malignancy of the Lung	17.4	8.7	22.4	16.7	19.7	6.6	Decreasing* .0005
Nonlung malignancy	17.4	16.5	15.7	23.8	5.0	6.9	Decreasing* .0009
Intubation	-	3.1	3.7	7.0	7.5	4.4	Increasing .22
CNS	5.8	7.9	15.7	2.4	7.9	2.8	Decreasing* .0002
Aortic/cardiac	3.5	6.3	1.5	-	4.3	0.6	Decreasing* .0004
Idiopathic	11.6	26.8	2.2	10.7	19.6	17.6	Increasing .53
Trauma	2.3	10.2	3.0	1.2	11.1	2.2	Decreasing* .01
Other	17.4	4.7	25.4	3.6	1.1	12.6	Decreasing .53

*Statistically significant, $P < .05$.
CNS = central nervous system.

Within the institution, similar changes were seen in bilateral vocal fold immobility. Surgical causes of bilateral vocal fold immobility have doubled, and malignant causes have declined. As with unilateral immobility, this may have resulted from increasing numbers of thyroidectomy, bilateral carotid endarterectomy, or other surgical procedures within the institution, but it is a departure from statistically significant trends in the literature, as shown in Table IV. Table IV demonstrates the changing trends in the etiology of bilateral vocal fold immobility. Iatrogenic injury from all surgical procedures is decreasing. However, nonthyroid surgeries, which are much less common causes, are increasingly responsible for bilateral immobility,[3,7–9] from 1.9% of bilateral immobility as published by Maisel and Ogura[3] in 1974 to 6.9% in the current study. The nonthyroid surgeries causing bilateral immobility are less often associated with carotid endarterectomy or anterior approaches to the cervical spine because these are typically one-sided procedures. Of the cases of documented bilateral immobility from nonthyroid surgery, two cases included carotid endarterectomy, and two included heart surgeries. The decline in total surgical causes of bilateral immobility in the literature may be a result of decreasing proportions of immobility from thyroidectomy. However, this study is consistent with early studies in which thyroidectomy accounted for a large majority of bilateral vocal fold immobility.[3,7] Similarly, malignancy caused less bilateral vocal fold immobility than expected by its increasing trend, which is possibly a consequence of the high number of iatrogenic injuries within this institution.

Of the 435 patients in the most recent cohort, there were many more cases of unilateral than bilateral immobility, which is expected because injury to one nerve is more likely than to both. In the current cohort, the left nerve (61%) was more commonly injured than the right (39%). This is consistent with the 1985 to 1995 study (left accounted for 62% and right 38%) as well as prior literature. This may be because the left recurrent laryngeal nerve is longer and more vulnerable. Specifically, it travels into the left chest and much of the difference can be accounted for by intrathoracic malignancy. In the most recent cohort, there were 38 cases of left vocal fold

TABLE IV.
Changing Etiology of Bilateral Vocal Fold Immobility.

	Maisel 1974, n = 54 (%)	Holinger 1976, n = 240 (%)	Feehery 1982–1997, n = 75 (%)	Benninger 1985–1995, n = 117 (%)	Henry Ford Hospital 1996–2005, n = 72 (%)	Trend *P* Value
Surgery (thyroid and nonthyroid)	42.6	58.8	19.0	25.7	55.5	Decreasing* .0015
Thyroid surgery	40.7	57.5	11.8	18.0	48.6	Decreasing* <.0001
Nonthyroid surgery	1.9	1.3	8.0	7.7	6.9	Increasing* .0015
Malignancy (lung and nonlung)	7.4	6.7	21.4	17.0	9.7	Increasing* .01
Intubation	1.8	—	9.3	15.4	9.7	Increasing .11
CNS	7.4	21.7	18.7	12.8	6.9	Decreasing .09
Idiopathic	3.7	3.3	13.3	12.8	8.3	Increasing* .002
Trauma	27.8	0.8	9.3	11.1	1.4	Decreasing .11
Other	9.3	8.8	9.3	5.1	8.4	Decreasing .45

*Statistically significant, *p* value <.05.
CNS = central nervous system.

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immobility secondary to malignancy and 22 (56%) were lung malignancies. There were 10 cases of right vocal fold immobility secondary to malignancy, two (20%) were of the lung and four (40%) were of the thyroid. There were 26 cases of right vocal fold immobility from thyroid surgery and 31 cases of left vocal fold immobility from thyroid surgery.

Accuracy of this study relies on adequate workup and documentation. In this study, etiology was properly evaluated based on the patient's history and physical examination. Many patients presented with a history of voice complaints following surgery, and therefore no additional studies were necessarily required. When immobility resulted from intubation, there was adequate workup to determine the pathophysiology of the immobility. An accurate account of how many patients had neuropathy versus ankylosis cannot be made in this retrospective study. Other patients had further diagnostic studies using imaging, such as a CXR, CT, MRI, or barium swallow, or EMG. These diagnostic tools are also valuable in development of a treatment plan. Frequently, patients presented to the office with previous workup, including outside imaging. One hundred percent of patients with a diagnosis of idiopathic etiology had additional workup, supporting adequate workup for that diagnosis.

Understanding the etiology of vocal fold immobility is an essential element of appropriate workup and management. How often iatrogenic injury causes permanent as opposed to temporary disability cannot be ascertained from this study. Of the 808 patients in the database with vocal fold immobility, 303 were excluded for inadequate documentation of immobility, etiology, or other necessary information for analysis. The documentation is often inadequate because patients were seen in the emergency room or office once and did not have follow-up with adequate workup. Many of these patients may have iatrogenic, either surgical or intubation-related immobility, or infectious etiologies, but were lost to follow-up, possibly because their symptoms resolved. This may be especially true of patients with unilateral vocal fold immobility. A prospective study would further delineate the prognosis and appropriate management for patients with a new onset vocal fold immobility.

The current study allows for appreciation of where to direct efforts to prevent neuropathy and minimize morbidity from vocal fold immobility. Intraoperative technique in neck surgery may particularly be an area to prevent injury, despite the low rates of permanent vocal fold immobility. Permanent recurrent laryngeal nerve injury reportedly occurs in about 1% of cases of thyroidectomy,[11] even with routine intraoperative identification of the recurrent laryngeal nerve.[12] The incidence of temporary recurrent laryngeal nerve injury varies from 2% to 12% depending on the indication for thyroidectomy.[11,12] Ipsilateral vocal fold immobility occurs in 2.6% to 6% of carotid endarterectomies,[13–15] with permanent paralysis in about 2% of cases.[15] Symptomatic recurrent laryngeal nerve injury occurs in about 2% of anterior cervical spine surgeries.[16] This prospective study showed an additional 11% of cases had asymptomatic immobility after 3 months.[16] Although the number of thyroidectomies performed within this institution may be increasing, the number of high-risk patients at this tertiary care facility in a large urban area with a large referral pattern may be disproportionate. Perhaps recurrent laryngeal nerve monitoring will show improved efficacy in the future, especially for seemingly high-risk patients, such as thyroidectomy patients with malignancies, large multinodular goiters, or previous thyroidectomy.

CONCLUSION

A review of 827 patients with vocal fold immobility (643 unilateral and 184 bilateral) over the last 20 years has shown consistency with previous reports of surgical iatrogenic injury causing increasing proportions of unilateral vocal fold immobility. Thyroid surgeries continue to cause a significant proportion of recurrent laryngeal nerve injury, but nonthyroid surgeries are increasingly responsible for such injuries. Carotid endarterectomy, anterior approaches to the cervical spine, and heart or great vessel surgeries currently cause more iatrogenic unilateral immobility than thyroid surgery does. However, this study shows some departure from trends identified in prior studies. Extralaryngeal malignancy, thought to be a growing cause of unilateral immobility, is a declining cause of unilateral immobility, for both lung and nonlung malignancies. Surgical injuries now surpass extralaryngeal malignancy as the most common etiology for unilateral immobility. Bilateral immobility, which is much less common than unilateral immobility is, has shown a different trend. Contrary to unilateral immobility, surgery is causing less bilateral immobility, although the trend is not strong and malignancy is increasingly responsible for bilateral immobility. Other common causes of vocal fold immobility, aside from surgery and malignancy, have remained stable. Idiopathic etiologies remain common causes of unilateral immobility and intubation remains a common cause of bilateral immobility.

Determining the mechanisms of vocal fold immobility plays a significant role in developing an appropriate management plan. Furthermore, understanding the etiology of vocal fold immobility within a population should allow for decreased morbidity and prevention of immobility.

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